CARL BRANDER

Zurich, CH
+41 76 474 75 55
carl.brander@gmail.com
linkedin.com/in/carl-brander

MSc. in Robotics from ETH Zurich with **excellent academic standing**.
Eight months of **Technical Program Management at Tesla** in Palo Alto, California.
Three months as **Digital Technology Consultant** at **McKinsey & Company.**
One year **leading a 38-person ETH engineering student project.**

Education

MSc. Robotics, Systems and Control | ETH Zurich **Sept. 2023 – Feb. 2026**
Ongoing Grade: 5.8/6

- Master Thesis: VLAs and LLMs with RL in mobile robotic grasping at the Microsoft Spatial & AI Lab
- CVPR25 Workshop Publication: "Reading in the Dark with Foveated Event Vision"
- Semester Thesis: "Text Reading with Event Cameras" at the Robotics and Perception Group with META Reality Labs
- Courses in AI, Computer Vision, Deep Learning, Autonomous Robots & Driving, Perception, AR & VR, Big Data

BSc. Mechanical Engineering | ETH Zurich **Sept. 2019 – Sept. 2023**
Grade: 5.6/6, top 3% of the Graduate Class

- SAS23 Publication: "Improving Data-Scarce Image Classification through Multimodal Synthetic Data Pretraining"
- Student Travel Grant Winner @ SAS23 Conference
- Bachelor Thesis: "Deep Sensor Fusion of Camera and ToF Sensors for Obstacle Detection with Nano UAV"
- Courses in Control Systems, Stochastics, Data & Algorithms, C++, Mechatronics, Thermo & Fluid Dynamics, Mechanics

Work Experience

Thesis Student | Microsoft - AI Lab **July 2025 – Feb. 2026**
Full-Time | Zurich **8 Months**

- Thesis Topic: VLAs and LLMs with RL in mobile robotic grasping with upcoming research publication

Digital Technology Consulting Intern | McKinsey & Company **April 2025 – June 2025**
Full-Time | Zurich and São Paulo **3 Months**

- Led and took full ownership over an end-to-end agile transformation for a 400-person technology division at one of Brazil's largest financial institutions, focusing on agile coaching, process redesign, and delivery acceleration.
- Led the design and analysis of a four-country market-growth survey for a top-10 global pharmaceutical company, building sophisticated Excel models that informed cross-country leadership of strategic growth insights.
- Received full-time return offer as both Digital and Integrative Fellow Consultant

Technical Program Management Intern | Tesla **Sept. 2023 – May 2024**
Full-Time | Palo Alto, California **8 Months**

- Took ownership of six low-voltage controller programs from scope definition and design to validation, leading multiple cross-functional teams to deliver functional products on time for vehicle development milestones.
- Consolidated inputs across Tesla into executive reviews, highlighting risks and opportunities to VPs and Directors.
- Used JIRA, ThinkCell, Confluence, and MS Office to manage, visualize, and communicate risks, increase efficiency, and unblock progress in time-critical programs (Optimus Humanoid Robot, Robotaxi, New Model 3 and Y, Cybertruck).
- Devised a workflow to improve controller end-of-line throughput by ~50% for the early-stage Optimus R&D Program.
- Received full-time return offer as TPM for both Optimus and vehicle teams

Project Lead | Swissloop **Sept. 2022 – Aug. 2023**
Full-Time | Zurich **12 Months**

- Led a multidisciplinary student project at ETH Zurich consisting of 38 students (12 working full-time), building a fully magnetically levitating Hyperloop prototype and winning the European Hyperloop week in Edinburgh, Scotland.
- Executing tight time schedules to meet engineering milestones, overseeing marketing, finances, engineering, and hiring, while creating tools to increase efficiency throughout the season and motivating the team.

Focus Coach | ETH Zurich **Sept. 2022 – July 2023**
Part-Time | Zurich 11 Months
- Coaching and grading eight focus students through their project year while setting them up for success.

Electrical Integration Engineer | Swissloop **Sept. 2021 – Aug. 2022**
Full-Time | Zurich 12 Months
- Working in a team of 12 full-time engineers developing a topology-optimized 3D-printed heatsink, compact subsystem integration, and cabling/EMC solutions to enable the development of a cargo-transporting Hyperloop prototype.

Academic Projects

VLAs with RL for grasping on mobile robots | Master Thesis, ETH Zurich
- Upcoming paper submission at the Thesis's end

Reading in the Dark with Foveated Event Vision | CVPR25 Workshop Publication
- First author paper publication based on Semester Thesis findings showcasing great improvement of an egocentric AI assistant reading text in challenging environments with an up to 2400x reduction in generated bandwidth.

Text Reading with Event Cameras | Semester Thesis, ETH Zurich
- Combining the Meta Project Aria glasses with an event camera to enable text recognition (OCR) in challenging low-light situations down to twilight brightness using a novel foveation approach, deep learning, and LLMs.

Improving Data-Scarce Image Classification Through Multimodal Synthetic Data Pretraining | SAS23 Publication
- Analyzing the performance of a self-developed, MM ToF, and RGB deep fusion model using custom synthetic data

Deep Sensor Fusion of Camera and ToF Sensors for Obstacle Detection with Nano UAV | Bachelor Thesis, ETH Zurich
- Developing a low-power deep sensor fusion network for ToF and RGB cameras for Object Detection on Nano UAVs.

Personal Projects

Solar Power Generation Forecasting using Local Weather Data for Home PV System
- Creating and deploying a NN on a self-hosted home server. Predicting solar energy production based on local weather forecast APIs. Enabling automatic solar-only energy surplus EV charging and visualization through a dashboard.

Smart Mirror Home Dashboard with Gesture Control and LLM Voice Assistant
- Using a Leapmotion Controller to gesture control a 45-inch self-built Smart Mirror as a Home Assistant control center.
- Integrating a "local detection" and open-source voice assistant satellite with an LLM fallback for smart home control.

Founded my own 3D Printing service company
- Established a 3D printing business producing high-quality parts for customers all over Switzerland for over 3 years.

Hobbies
- Road biking, fitness, playing tennis, as well as 3D printing, smart home automation, DIY electronics, and self-hosting.
- 3rd Kyu (brown belt) in Shotokan Karate

Skills

Technical
- Programming: Python, PyTorch, (ROS2), C++, LaTeX, (C, Matlab, SQL, NoSQL)
- Simulation & Design: SiemensNX, (Enovia, FloEFD)
- General: Docker, Linux, Proxmox, Unity

Operational
- Software: MS Office (especially Excel), Jira, ThinkCell, Confluence, Trello
- Soft-Skills: Stakeholder Management, Time-Management, and strict prioritization

Languages
- German Mother-Tongue
- Swiss-German Mother-Tongue
- English C2 (Proficient)
- French: B2 (Intermediate)